Windows Security Event Log Analysis

Project Type: Beginner SOC Analyst Practical Project
Tool Used: Windows Event Viewer
Log Source: Windows Security Logs

1. Introduction

Windows Security Event Logs record security-related activities on a Windows computer, including login attempts, account activity, and other security events.

In this project, I used Windows Event Viewer to examine selected Security Log entries. The investigation focused on Event ID 4625 (failed logon) and Event ID 4624 (successful logon) to understand authentication activity and identify the information available for security analysis.

2. Objectives

- Understand how to access Windows Security Logs.
- Identify failed and successful logon events.
- Examine Event IDs 4624 and 4625.
- Understand logon types, failure codes, and process information.
- Document findings without assuming that an event is malicious.
- Identify useful security monitoring recommendations.

3. Tools Used

- Windows Event Viewer
- Windows Security Logs

4. Methodology

1. Opened Windows Event Viewer.
2. Navigated to **Windows Logs → Security**.
3. Located and examined an Event ID 4625 record.
4. Reviewed its failure information, logon type, process information, and network information.
5. Located and examined an Event ID 4624 record.
6. Reviewed its account, logon type, process information, and network information.
7. Compared the two selected records and documented the findings.

5. Investigation 1 — Failed Logon (Event ID 4625)

Event ID: 4625
Event description: An account failed to log on.

Logon Type: 2 — Interactive logon
Failure Reason: Unknown user name or bad password
Status: 0xC000006D
Sub Status: 0xC000006A
Caller Process: msedgewebview2.exe
Source Network Address: Not available in the examined record

Analysis

Event ID 4625 indicates that a logon attempt failed.

The status code 0xC000006D indicates a general logon failure. The substatus code 0xC000006A indicates that the password was incorrect.

The event shows Logon Type 2, which represents an interactive logon request. The caller process was msedgewebview2.exe, a Microsoft Edge WebView2 process.

The source network address and workstation name were not populated in the examined event. Therefore, the source of the request cannot be identified from this record alone.

Finding: One failed logon event was examined. The record indicates an authentication failure, but the available information is not sufficient to conclude that it was caused by an attacker.

6. Investigation 2 — Successful Logon (Event ID 4624)

Event ID: 4624
Event description: An account was successfully logged on.
Account: SYSTEM
Account Domain: NT AUTHORITY
Logon Type: 5 — Service logon
Process Name: C:\Windows\System32\services.exe
Source Network Address: Not available in the examined record

Analysis

Event ID 4624 indicates that a logon session was successfully created.

The examined record shows the SYSTEM account and Logon Type 5, which represents a service logon. The process shown is services.exe, the Windows Service Control Manager.

This information is consistent with Windows service activity. It is not evidence of a person successfully logging in. The examined record does not contain a source network address.

Finding: The selected successful logon record is consistent with a Windows service logon. No evidence in this record alone establishes malicious activity.

7. Comparison of the Examined Events

Detail	Event ID 4625	Event ID 4624
Meaning	Failed logon	Successful logon

Logon Type	2 — Interactive	5 — Service
Account information	Target account not populated	SYSTEM
Process	msedgewebview2.exe	services.exe
Source IP	Not available	Not available

Important note: These are two separately examined event records. Their timestamps and account details have not been established as matching, so this investigation does not conclude that the successful logon was related to the failed logon.

8. Key Findings

- Examined Windows Security Event ID 4625 to understand a failed authentication attempt.
- Identified substatus 0xC000006A, indicating an incorrect password.
- Examined Event ID 4624 and identified a Logon Type 5 service logon.
- Observed that the source network address was unavailable in both selected records.
- Identified msedgewebview2.exe and services.exe in the process information.
- Did not find enough evidence in these individual records to classify either event as malicious.

9. Security Recommendations

- Monitor repeated failed logon events, especially when they occur in a short period.
- Review Event IDs 4624 and 4625 regularly.
- Investigate unusual login times, unfamiliar accounts, and unexpected logon types.
- Correlate events using timestamps, account details, host information, and source addresses where available.
- Use additional logs and endpoint information before deciding whether an event is malicious.

10. Conclusion

This project provided practical experience using Windows Event Viewer to examine authentication-related Security Logs.

By analysing selected Event IDs 4624 and 4625, I learned how to distinguish failed logon attempts from successful service logons and how to interpret logon types, status codes, and process information.

The investigation also demonstrated the importance of using available evidence carefully and avoiding unsupported conclusions when investigating security events.

11. Evidence

Add your own screenshots here if available:

- Screenshot 1: Windows Event Viewer with Security Logs open.
- Screenshot 2: Event ID 4625 details.
- Screenshot 3: Event ID 4624 details.

If screenshots are unavailable, do not insert unrelated or sample images. The written investigation findings document the records examined.